

# Daily Sequences Drill #3

Sheet by David Akinyele-Aje

| Section | Topic                    | Questions | Target time  |
|---------|--------------------------|-----------|--------------|
| A       | Rapid Recognition        | 10        | 2 min 30 sec |
| B       | Manipulation Drills      | 10        | 8 min        |
| C       | Substitution & Structure | 8         | 10 min       |
| D       | Challenge                | 5         | 15 min       |

New toolkit today: Geometric Sequences & Series · Sum to Infinity · The Convergence Condition  $|r| < 1$ .

Attempt each question before checking answers. Time yourself. No calculators.

## Section A Rapid Recognition

(≤15 s each)

Ratios, terms, and the sum-to-infinity formula on sight.

- Write down the common ratio of the GP 3, 6, 12, 24, ...
- A GP has first term 5 and common ratio 2. Find the 6th term.
- A GP has first term 2 and common ratio 3. Find  $S_5$ .
- State the condition on  $r$  for an infinite GP's sum to converge.
- A GP has first term 8 and common ratio  $\frac{1}{2}$ . Find its sum to infinity.
- Which of these three sequences is geometric: (i) 2, 4, 6, 8    (ii) 2, 4, 8, 16    (iii) 2, 4, 7, 11?
- Write down the sum-to-infinity formula for a GP with first term  $a$  and ratio  $r$  (where  $|r| < 1$ ).
- True or false: a GP with common ratio  $r = 1$  has a well-defined sum to infinity.
- A GP has first term 100 and common ratio  $-\frac{1}{2}$ . Find its sum to infinity.
- Write the next term of the GP 81, 27, 9, 3, ...

## Section B Manipulation Drills

(≤50 s each)

Solve for the GP's parameters, then use them — always check convergence first.

- A GP has first term 4 and common ratio 3. Find the smallest  $n$  for which  $S_n > 1000$ .
- For which values of  $r$  does the infinite GP with first term 5 have a finite sum?
  - all real  $r$
  - $r > 0$  only
  - $-1 < r < 1$

- D)  $r \neq 1$
3. A GP has first term 3 and common ratio  $\frac{2}{3}$ . Find its sum to infinity.
4. A GP has sum to infinity 20 and first term 5. Find  $r$ .
- A)  $\frac{3}{4}$   
B)  $\frac{1}{4}$   
C)  $\frac{4}{5}$   
D)  $-\frac{3}{4}$
5. A GP has first term 1 and sum to infinity 2. Find the second term.
- A)  $\frac{1}{2}$   
B)  $\frac{1}{4}$   
C) 2  
D) 1
6. An AP and a GP both have first term 4. The AP has common difference 2; the GP has common ratio 2. Which sequence has the larger 10th term?
- A) The AP  
B) The GP  
C) They are equal  
D) Cannot be determined
7. A GP has all terms positive and common ratio  $r > 1$ . Does its sum to infinity exist?
- A) Yes, always.  
B) No, never — it diverges.  
C) Only if the first term is small.  
D) Only if  $r < 2$ .
8. Express the recurring decimal  $0.\overline{3} = 0.333\dots$  as a fraction, using the geometric series sum-to-infinity formula.
9. A GP has first term 2 and common ratio  $-3$ . Find  $S_4$ .
- A) 40  
B)  $-40$   
C) 160  
D)  $-160$

10. Recall from Day 2 that  $(1 - x)^{-1} = 1 + x + x^2 + \cdots$  for  $|x| < 1$ . Using  $x = \frac{1}{3}$ , what does this series sum to?

- A)  $\frac{3}{2}$
- B)  $\frac{2}{3}$
- C) 3
- D) 2

### Section C Substitution & Structure

(≤90 s each)

*Convergence first, computation second.*

1. A GP has first term 1 and unknown common ratio  $r$ . Its sum to infinity exists and equals 4. What must be true about  $r$ ?

- A)  $r = \frac{3}{4}$  exactly.
- B)  $-1 < r < 1$ , but no exact value is determinable.
- C)  $r = \frac{4}{3}$ .
- D) Infinitely many values of  $r$  satisfy this.

2. A geometric sequence has first term  $a$  and common ratio  $r$ , both positive integers with  $r > 1$ . Let  $S_n$  denote the sum of the first  $n$  terms. Given  $S_{18} - S_{12} = kS_6$  for some positive integer  $k$ , find the smallest possible value of  $k$ . (after TMUA 2022 Paper 1 Q8)

- A)  $2^6$
- B)  $2^{12}$
- C)  $2^{18}$
- D)  $\frac{2^6}{2^6 - 1}$
- E)  $2^6(2^6 - 1)$

3. An infinite geometric series has first term  $a = 6$  and common ratio  $r = \frac{3k}{10}$ , where  $k$  is chosen uniformly at random from the integers  $-5, -4, \dots, 4, 5$  (11 equally likely values). Find the probability that the series converges *and* its sum to infinity exceeds 5. (after TMUA 2023 Paper 1 Q18)

- A)  $\frac{4}{11}$
- B)  $\frac{5}{11}$
- C)  $\frac{7}{11}$
- D)  $\frac{3}{11}$

4. The series  $\sum_{k=1}^{\infty} ar^{k-1}$  converges (for real  $a, r$ ) when  $|r| < 1$ . If  $a$  and  $r$  are instead allowed to be complex numbers, which is the correct convergence condition?

- A)  $|r| < 1$  (the same condition, using the complex magnitude).
- B)  $r < 1$  (a real inequality).

- C)  $a < 1$ .
- D) There is no meaningful convergence condition for complex  $r$ .
5. Convert the recurring base-2 number  $0.\overline{0111} = 0.0111\,0111\,0111\ldots$  (base 2) to a fraction in base 10. (after TMUA 2023 Paper 2 Q15)
- A)  $\frac{1}{3}$
- B)  $\frac{1}{5}$
- C)  $\frac{7}{15}$
- D)  $\frac{7}{16}$
6. Which of the following statements about arithmetic and geometric series is FALSE?
- A) An AP's terms grow linearly in  $n$ ; a GP's terms with  $|r| > 1$  grow exponentially in  $n$ .
- B) An AP diverges to  $\pm\infty$  (or stays constant) as  $n \rightarrow \infty$ , unless  $d = 0$ .
- C) A GP can converge to a finite sum to infinity even though no individual term is exactly zero, provided  $|r| < 1$ .
- D) Every convergent GP has all its terms equal to zero eventually.
7. Using  $(1-x)^{-1} = \sum_{k=0}^{\infty} x^k$  from Day 2, evaluate  $\sum_{k=0}^{\infty} \frac{1}{5^k}$  by choosing an appropriate value of  $x$ .
- A)  $\frac{5}{4}$
- B)  $\frac{4}{5}$
- C) 5
- D)  $\frac{1}{4}$
8. Let  $S = 1 + 2x + 3x^2 + 4x^3 + \cdots$ . Recall from Day 2 that this is exactly  $(1-x)^{-2} = \sum_{k=0}^{\infty} (k+1)x^k$ . Evaluate  $S$  at  $x = \frac{1}{4}$ .
- A)  $\frac{16}{9}$
- B)  $\frac{4}{3}$
- C)  $\frac{9}{16}$
- D)  $\frac{3}{4}$

**Section D Challenge**

(TMUA / SMC difficulty)

*Insight over computation. Each question has a short, elegant route — find it.*

1. Prove that  $\sum_{k=1}^{\infty} \frac{1}{2^k} = 1$  exactly (not merely “approaches 1”), using the sum-to-infinity formula.
- Then explain why the partial sum  $S_n = \sum_{k=1}^n \frac{1}{2^k} = 1 - \frac{1}{2^n}$  never actually equals 1 for any finite  $n$ .

2. Let  $x = 0.\bar{9} = 0.999\dots$  (a recurring 9). Writing  $x$  as an infinite geometric series and summing it, determine  $x$  exactly.
- A)  $x = 1$  exactly.
  - B)  $x$  is infinitesimally less than 1 but not equal to it.
  - C)  $0.\bar{9}$  is undefined.
  - D)  $x = \frac{9}{10}$ .
3. A ball is dropped from height  $h$ . Each time it bounces, it rebounds to a fraction  $r$  of its previous height ( $0 < r < 1$ ). Find, in terms of  $h$  and  $r$ , the total distance the ball travels (up and down, forever) before coming to rest.
- A)  $h \cdot \frac{1+r}{1-r}$
  - B)  $h \cdot \frac{2}{1-r}$
  - C)  $h \cdot \frac{1}{1-r}$
  - D)  $h(1+2r)$
4. The sum to infinity of a GP equals the sum of its first two terms. What can be said about the common ratio  $r$ ?
- A)  $r = 0$  (the “GP” is actually constant).
  - B)  $r = \pm 1$ .
  - C)  $r = \frac{1}{2}$ .
  - D) No such  $r$  exists.
5. A sequence is defined by  $a_1 = 3$  and  $a_{n+1} = \frac{1}{2}a_n$  for  $n \geq 1$ . Using the sum-to-infinity formula, evaluate  $\sum_{n=1}^{\infty} a_n$ .
- A) 6
  - B) 3
  - C) 4.5
  - D) It cannot be summed, since it's defined by a recurrence, not a formula.

---

*“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”*

---

Found a mistake or want to contribute a sheet?

Visit the open-source project at [github.com/The-CerealDev/speed-maths](https://github.com/The-CerealDev/speed-maths)